QUANTUM ENTANGLEMENT

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Quantum entanglement is a phenomenon in quantum mechanics where quantum states of two or more objects are interconnected such that the quantum state of each object cannot be described independently. This creates correlations between observable physical properties that appear to defy classical intuitions about the separability of distant objects. The phenomenon was the subject of a 1935 paper by Albert Einstein, Boris Podolsky, and Nathan Rosen, and several papers by Erwin Schrödinger shortly thereafter, describing what came to be known as the EPR paradox. Einstein and others considered such behavior impossible because it violated the local realist view of causality and argued that the accepted formulation of quantum mechanics must therefore be incomplete. Later, however, the counterintuitive predictions of quantum mechanics were verified experimentally. Entanglement has been demonstrated experimentally with photons, neutrinos, electrons, molecules as large as buckyballs, and even small diamonds. The utilization of entanglement in communication, computation and quantum radar is a very active area of research and development. Quantum entanglement is the basis for emerging technologies such as quantum computing and quantum cryptography, and has been used to realize quantum teleportation experimentally. At the same time, it prompts some of the more philosophically oriented discussions concerning quantum theory. The nature of quantum entanglement challenged Einstein's principle of local realism, leading to his famous comment that entanglement involved 'spooky action at a distance'. The EPR paper generated significant interest among physicists, who were divided about whether the authors had proved that quantum mechanics was incomplete. Bell's theorem proved that quantum physics is incompatible with local hidden-variable theories. John Stewart Bell showed that correlations between measurements on entangled particles are stronger than any classical system could exhibit. These predictions were later confirmed in experiments, showing that quantum mechanics accurately describes reality.